

Unit 6 – Inference for Categorical Data: Proportions

Notes 5 – Errors and Power

The Courtroom Analogy

		The Truth	
Jury's Decision		Not Guilty	Guilty
	Guilty		
	Not Guilty		



What would be considered the "worse" mistake?

A jury is going to decide whether the defendant is guilty or not guilty



Always start with the assumption that the defendant is innocent



The court presents evidence and a decision is made



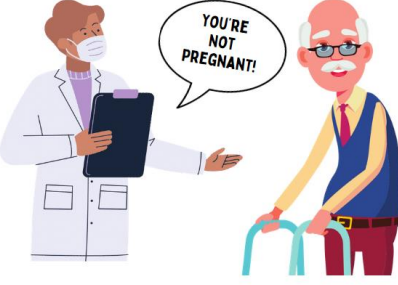
		The Truth About the Population	
Your Decision			

Example:

H_0 : You are not pregnant

H_A : You are pregnant

When you go into the doctor's office, it is always assumed you are not pregnant until we get tests that tell us otherwise.

The Truth About the Population		
Your Decision		
	H_0	H_a
	H_a	H_0
	Truth: Decision: 	Truth: Decision: 
	Truth: Decision: 	Truth: Decision: 

Type I Error	Type II Error	Power
• The probability of a Type I error is equal to the • $P(\text{Type I Error}) = \underline{\hspace{2cm}}$ • Our significance level tells us what p-value is "low enough"	• $P(\text{Type II Error}) = \underline{\hspace{2cm}}$	• The probability we reject the null correctly is 1 minus the probability we reject the null incorrectly • Power = $\underline{\hspace{2cm}}$

Errors and Their Relationships

- Type I and Type II errors have an _____ relationship: as the probability of one increases, the probability of the other decreases
- Our Type I error is set with our _____
- The higher our significance level is, the lower our _____ becomes, which is why α and β have an indirect relationship
- The higher our significance level, the more likely we will _____, which increases the likelihood we do that incorrectly (as well as correctly)
- Therefore, Type I error and Power have a _____ relationship: as the probability of Type I Error increases, the higher the Power of the test

Interpreting Errors: An Example

- What type of error would you rather have? Well that depends on the problem!
- For some tests, we want a low Type I but for others we want a low Type II.
- This is why being able to interpret errors is a key skill on the AP Exam.

A drug manufacturer claims that less than 10% of patients who take its new drug for treating gestational diabetes will experience nausea. To test this claim, researchers conduct an experiment at the 5% significance level.

a) What are the null and alternative hypothesis?

b) Describe a Type I error and a Type II error in this setting, and explain the consequences of each.

c) The test has a power of 0.54 to detect that the alternative is true. Explain what the power means in this setting. What is one way we can increase the power?

d) You know that the power of this test at the 5% significance level against the alternative is 0.54. If you decide to use $\alpha = 0.01$ instead of the 5% significance level, with no other changes to the test, will the power increase or decrease? Explain.